How to Make Work Meaningful?*

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Abstract

Many people derive a sense of purpose from their job – they consider work to be a source of *meaning*. But how to make work meaningful? The theoretical literature proposes a four factor model based on psychological self-determination theory and pro-social work motivation. We empirically assess this theory and estimate a nonlinear production function for work meaning. We find that all four pathways can generate meaning, and document substantial complementarities. The magnitude of these results is sizeable, since we estimate the compensating differential for a standard deviation increase in work meaning to be over 500 dollars of monthly wages.

Keywords: work meaning, production function, factor model, measurement error

JEL Codes: D91, J32, D24

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1 Introduction

> *"If one wanted to crush and destroy a man entirely [...], make him do work that is completely and utterly devoid of usefulness and meaning."*
>
> Fyodor Dostoevsky (1860), The House of the Dead

Many people derive a sense of impact and purpose from their job – they consider work to be a source of *meaning*. A substantial body of literature in psychology, sociology, and economics has studied the importance of meaningful work (see Rosso et al. (2010), Martela and Riekki (2018), and Rosso et al. (2010) for recent reviews). These studies find that working a meaningful job is related to a myriad of benefits in the workplace, such as higher productivity (Ariely et al., 2008), less turnover (Burbano et al., 2023), lower absenteeism (Nikolova and Cnossen, 2020), and lower reservation wages (Hu and Hirsh, 2017; Kesternich et al., 2021). Beyond the workplace, meaningful work has been linked to lower levels of psychological distress (Steger et al., 2012) and higher overall well-being (Arnold et al., 2007).

So how can we make work meaningful? Theoretical work in psychology and economics has identified four distinct *pathways* to meaning (Martela and Riekki, 2018; Cassar and Meier, 2018). Three of them are derived from psychological self-determination theory, which posits that human well-being is rooted in the satisfaction of three basic needs (see Ryan and Deci (2000) and Deci and Ryan (2013)). The first need is that for *autonomy*, or a sense of freedom and flexibility in deciding one's work situation. The second need is *competence*, which refers to people's ability to apply their talents, skills, and knowledge in their job. The third one is *relatedness*, which captures the connection people have with their colleagues, supervisors, and the company. The final pathway to work meaning is derived from the the literature on pro-social behavior (see Dunn et al. (2008)). The sense of having a pro-social impact, or *beneficence*, can make work more meaningful insofar as it gives individuals the feeling that they are making a positive contribution to society and helping other people.[1]

[1]When we talk about work meaning – or any of the pathways – throughout this paper, we are referring to the individual's own perception of being engaged in meaningful work or about their perceived level of autonomy. Two individuals that work the same jobs may perceive their levels of autonomy, and thus likely their levels of meaning, to be very different, depending on their idiosyncratic assessment of their working conditions. That being said, objective measures of these dimensions can be informative, as they are likely to be highly correlated with perceived levels.

Recent studies have shown that the different pathways correlate significantly with self-reported levels of work meaning (see Martela and Riekki (2018), Nikolova and Cnossen (2020), and Burbano et al. (2023)). To estimate the *relative importance* of these pathways in making work meaningful, two problems need to addressed. The first one is that neither work meaning nor the pathways are easily measurable. The survey questions that are typically used are highly subjective, which means that respondents are likely to differ in both their interpretation of the questions and the scales of the answers. This can lead to substantial measurement error that causes non-negligible attenuation bias in the estimated parameters. The second problem is that individuals select into jobs with specific combinations of the different pathways. This can be problematic if unobserved characteristics, such as productivity, personality traits, or social norms determine both an individual's (perception of) work meaning and the different pathways.

This paper exploits rich panel data from the American Working Conditions Survey (AWCS) to overcome these challenges.[2] To address concerns of measurement error, we introduce a measurement system that combines several potentially noisy measures of work meaning, autonomy, beneficence, competence, and relatedness. The factors constructed through the measurement model are then used as inputs in a flexible nonlinear production function of work meaning. We then introduce individual fixed effects to the production function to estimate how work meaning is produced *within* each individual. This addresses concerns related to idiosyncratic interpretation of the question and other time-invariant individual heterogeneity.

The first main result in this paper is that work meaning can be created through all four pathways. In our preferred specification, the elasticity of work meaning with respect to all pathways is positive and statistically significant. We find that the most effective pathway to work meaning is competence. The direct output elasticity of work meaning with respect to competence, holding all other pathways constant, is 0.75. The next most effective pathways are beneficence, with a direct output elasticity of 0.37, and autonomy, with an elasticity of 0.28. The effect of relatedness is also positive but smaller, with an elasticity of 0.09. There is no direct comparison for these results, but we find them to be somewhat in line with previous work by Martela and Riekki (2018) and Nikolova and Cnossen (2020). The main difference is that we find a substantially larger effect

[2]The American Working Conditions Survey (AWCS) is conducted several times through the American Life Panel (ALP), a representative survey of workers in the United States. The data is collected by the RAND Corporation.

for competence, which may be due to cultural differences or attenuation bias.

The second main result is that we find interactions between the different pathways to be of first order importance. For example, we find that autonomy and competence are complementary, with a coefficient of 0.43 on their interaction in our preferred specification. This is larger than all direct effects except for that of competence. This result is furthermore in line with long standing predictions from psychological theory by Ryan and Deci (2000). The interpretation is straightforward – individuals need some degree of autonomy to express their competence, or conversely may benefit from more structure when feelings of competence are low. We also find that competence and relatedness act as complements – suggesting that people especially derive meaning from feeling competent when working with colleagues whom they care about – but this coefficient is smaller and statistically not significant.

We also want to highlight the importance of allowing for *negative* interactions between the pathways. In our preferred specification we document significant negative interactions between two combinations of pathways. The largest interaction is that between autonomy and relatedness. We believe that people at the lower of for both pathways drive these results. These individuals have little autonomy over whom they work with or whom they take orders from. This can have a particularly negative impact on work meaning when they do not at all relate to their colleagues or superiors. This finding is furthermore in line with some influential psychological theories such as Steinberg and Silverberg (1986), but not necessarily with self determination theory as proposed in Ryan and Deci (2000). The second negative relation is between relatedness and beneficence. We interpret this correlation in terms of individuals that work jobs with a high pro-social dimension, such as nurses, teachers, or therapists, having sufficient social interactions during their daily activities, and not deriving much further meaning from relating to their colleagues. On the other hand, individuals that do not relate at all to their colleagues and do not work a highly pro-social job could become isolated which can be particularly harmful for their work meaning.

We further interpret the magnitude of these results by translating them into a money metric. We do so by estimating the equilibrium price for work meaning in the labor market, and then calculating how much money changes in the different pathways would generate. We find that work meaning is valuable, as the compensating differential for a standard deviation increase in log meaning is more than 500 dollars of monthly wages. This estimate implies that a unit increase in both log competence and autonomy would on average generate the equivalent of more than 600 dollars of monthly wages. We find comparable but slightly lower values for a joint increase in beneficence and competence, relatedness and competence, and autonomy and beneficence.

The final step in our analysis considers in which dimensions different occupations have room for improvement, and how much work meaning these improvements could generate. This highlights the immediate practical implications of our results for firms and organizations that could consider offering jobs with more meaning to attract – or retain – more employees. The overall most effective way to do so in most occupations is fostering workers' feelings of competence. Recognizing their personal and social contributions has been highlighted as an important mechanism (see Cassar and Meier (2018) and Ellingsen and Johannesson (2007)). This leaves room for a substantial impact of effective management practices. Work in psychology such as Bauer and Mulder (2006) has shown that implementing procedures that allow employees to provide upward feedback may be an effective way to increase feelings of competence.

The second way to increase work meaning among employees is by attempting to improve employees' feelings of beneficence. This would be a particularly promising direction in technical occupations – such as lawyers, engineers, and computer scientists – which already score high on all three psychological needs, where further improvement may be difficult. The substantial efforts that companies have been devoting to large scale corporate social responsibility programs and carefully crafted mission statements can be considered an important first step in this direction (see Bhattacharya et al. (2008) and Cassar and Meier (2018)). The opposite is true for workers in health- and education related occupations, who report high levels of beneficence but work jobs that are designed substantially worse, particularly in terms of autonomy. There is furthermore increasing evidence that levels of autonomy have been decreasing in these occupations (see Taylor et al. (2002) and Petrakaki and Kornelakis (2016)) and in general throughout European countries (see Green (2006) and Lopes et al. (2014)). The additional further decrease in autonomy associated with robotization documented in Nikolova et al. (2024) suggests that policy efforts will be necessary.

The remainder of this paper is organized as follows. We first present the production model in section 2. Section 3 discusses the empirical strategy to estimate the model. Section 4 introduces the data and discusses the selected measures. We then present and discuss the results in section 5, before concluding in section 6.

2 Model

We build on a rich literature in psychology and economics that argues work meaning can be created through four pathways (see Martela and Riekki (2018) and Cassar and Meier (2018)). The first three are derived from psychological self determination theory, one of the workhorse models in psychology to explain human motivation and well being (Ryan and Deci, 2000, 2017). Self determination theory posits that human well being is rooted in the satisfaction of three basic psychological needs. The first need is one for *autonomy*, which refers to the degree to which someone perceives to be in control of initiating and sustaining their own actions. The second need is for *competence*, which relates to feelings of being capable and efficient in one's work tasks, but also of being able to accomplish projects and achieve goals. The final need is for *relatedness*, which captures the interpersonal dimension and connection to one's employer. This particularly captures the feeling of being connected to work relationships and general feelings of belonging to a community in the workplace. The final pathway to work meaning – *beneficence* – is derived from the literature on pro-social work motivation (e.g. Dunn et al. (2008)) and reflects the satisfaction that comes from doing work that is beneficial for society or that helps other people.

The idea of modeling work meaning as a production process with the four pathways as inputs stems from Cassar and Meier (2018). The production model that we estimate builds on their insights but differs because we abstract from the effort choice, which is more in line with typical models of meaning in the psychological literature (e.g. Rosso et al. (2010) and Martela and Riekki (2018)). We write the production function as:

$$M_i = f(A_i, C_i, R_i, B_i, \eta_i), \tag{1}$$

where M represents the experienced level of *work meaning* for individual i, and A, C, R and B

respectively denote their experienced *autonomy*, *competence*, *relatedness* and *beneficence*. The term η_i represents an idiosyncratic productivity shock that captures individual specific effects and omitted inputs.

We parameterize the production process through a trans-log function, which allows for flexible substitution patterns between the different pathways. This is guided by theoretical work on self determination theory that goes back to at least Ryan and Deci (2000), who argue that interactions between the pathways are important. The benefits of this functional form over the commonly used (nested) Constant Elasticity of Substitution (CES) functions are that it does not impose prior restrictions on the nature of substitution patterns, which are allowed to be both positive and negative (see Agostinelli and Wiswall (forthcoming)). The production function equation (1) becomes:

$$\ln M_i = \sum_{P \in \mathcal{P}} \gamma_P \ln(P_i) + \sum_{P \in \mathcal{P}} \sum_{\substack{P' \in \mathcal{P} \\ P' \neq P}} \gamma_{PP'} \ln(P_i) \ln(P'_i) + \eta_i, \tag{2}$$

where $\mathcal{P} = \{A, C, R, B\}$ denotes the set of pathways. The γ_P coefficients represent the direct effect output elasticity of work meaning with respect to pathway P, and the $\gamma_{PP'}$ coefficients on the interaction terms represent pair-specific complementarities between pathways P and P'.

Empirical Approach. Two problems need to be addressed when estimating equation (2). First, the variables in our model – work meaning and the pathways – are hard concepts to measure. Different measures and procedures to aggregate them have been considered in the literature, such as Nikolova and Cnossen (2020), who run independent principal component analyses for each pathway, and Burbano et al. (2023), who take averages of different measures. We formalize their approaches and model measurement error through a measurement system, which combines several imperfect measures that may differ in location, scale, and information content into novel factors.[3] The estimator we use in this paper has recently been introduced by Agostinelli and Wiswall (forthcoming), and consists of instrumenting measures in the model with alternative omitted measures.

The second problem is that the error terms in the production function may not be exoge-

[3]This is the predominant approach in the literature on human capital formation – see Cunha and Heckman (2007), Cunha et al. (2010), and Agostinelli and Wiswall (forthcoming).

nous, and are likely to be correlated with both the inputs of the production function and work meaning. This could be the case when idiosyncratic factors, such as an individual's personality traits or productivity, determine both the job an they sort into and their (perceived) level of work meaning. We take an example from the literature on compensating differentials (see a recent review in Lavetti (2023)). Suppose that the perceived subjective levels of the pathways to work meaning are correlated with a latent objective level. More productive individuals are likely to sort into jobs that are better in all these latent objective dimensions and in several other characteristics. Some of these other characteristics could generate meaning for some individuals but are omitted in our production function, introducing a spurious correlation between work meaning and the pathways that is driven by productivity differences. To address this problem and any other concern related to unobserved time-invariant individual heterogeneity, such as the interpretation of the questions and scaled of the answers, we introduce individual fixed effects into the production function.[4] The following sections present a more detailed overview of our estimation strategy.

3 Empirical Strategy

3.1 Measurement Model

We assume that the different observed measures and latent factors are related through a log-linear measurement model with the following structure[5]:

$$Q_j^F = \mu_j^F + \lambda_j^F \ln F + \psi_j^F \text{ for all } F \in \{\mathcal{P}, M\}, \tag{3}$$

where Q_j^F is the j-th measure of the latent factor F, μ_j^F is the location of the measure, λ_j^F the factor loading of the j-th measure on the latent variable, and ψ_j^F a measure-specific error term. We denote the set of all latent factors (that is, *autonomy*, *competence*, *relatedness*, *beneficence*

[4]The latter is a problem that is similar to the a well-known issue in the literature on the determinants of happiness (see (Ferrer-i Carbonell and Frijters, 2004))

[5]This assumption is clearly not without loss of generality but is prevalent in most empirical models (see e.g. Attanasio et al. (2020)), as pointed out in Agostinelli and Wiswall (forthcoming). It is well known that the latent distributions of each factor can be identified with three or more dedicated measures – see for example Cunha et al. (2010).

and *work meaning*) by $\{\mathcal{P}, M\}$ and assume that the measurement equations follow a similar structure. The error terms are assumed to be mean zero and independent of each other, the latent factors, and the production shocks η_i.

We normalize the loading of the first measure of each latent factor (λ_1^F) to unity without loss of generality. Additionally, we normalize the logarithm of all latent factors $\ln(F)$ to be mean zero. Given these normalizations, we can retrieve the factor loadings by taking the sample analog of:

$$\lambda_j^F = \frac{\text{Cov}(Q_j^F, Q_{j'}^F)}{\text{Cov}(Q_1^F, Q_{j'}^F)} \text{ for all } j \neq j' \text{ and } F \in \{\mathcal{P}, \mathcal{M}\}, \tag{4}$$

$$\mu_j^F = \mathbb{E}(Q_j^F) \text{ for all } j \text{ and } F \in \{\mathcal{P}, M\}. \tag{5}$$

We then use these estimates to construct the following residual measures of the latent variables:

$$\hat{F}_j = \frac{Q_j^F - \mu_j^F}{\lambda_j^F} \text{ for all } j \text{ and } F \in \{\mathcal{P}, M\},$$

which we use to estimate the production function. Notice that for these measures the following equality holds:

$$\ln F + \psi_j^F = \frac{Q_j^F - \mu_j^F}{\lambda_j^F} = \hat{F}_j, \text{ for all } F \in \{\mathcal{P}, M\}. \tag{6}$$

3.2 Estimation of the Production Function

We rewrite our production function by substituting in the residual measures defined through equation (6), which after some rewriting (details in Appendix A.1) yields:

$$\hat{M}_{ij} = \sum_{P \in \mathcal{P}} \gamma_P \hat{P}_{ij} + \sum_{P \in \mathcal{P}} \sum_{\substack{P' \in \mathcal{P} \\ P' \neq P}} \gamma_{PP'} \hat{P}_{ij} \cdot \hat{P}'_{ij} + \phi_i + \phi_o + \xi_{ij}(\hat{P}_{ij}, \hat{P}'_{ij}). \tag{7}$$

We can see that the error term in this equation (ξ_{ij}, see Appendix A.1 for its full expression) is correlated with the pathways, so we cannot directly estimate this function with an ordinary least squares regression. However, as noted in Agostinelli and Wiswall (forthcoming), alternative omitted measures of the latent factors are valid instruments under the assumptions of the measurement model. They are relevant by definition, and uncorrelated with all components of

ξ_{ij} under the measurement model assumptions.

We address potential correlation between the measures and the latent factors due to (time invariant) idiosyncratic factors by introducing individual fixed effects (ϕ_i). As discussed in section A.1 this addresses concerns of differences in productivity or personality traits that cause individuals to sort into specific occupations and deals with the idiosyncrasies in interpretation of the survey. We also introduce fixed effects for occupations (ϕ_o) to account for any occupation-specific generation of work meaning. The magnitude of these coefficients also gives us an indication as to how much of the production of work meaning our model captures, because large occupation fixed effects suggest that there are important characteristics of some occupations that lead to higher (or lower) work meaning that are not captured by our model. See Appendix A.6 for a more detailed description of the estimation procedure.

Given the two-step nature of the estimation strategy, we rely on a bootstrapping procedure for inference. Concretely, we estimate both the measurement system and the production on 1.000 bootstrap samples, which are all of the same size as the original sample and draw with replacement at the individual-level.

4 Data

We estimate our model on data from the American Working Conditions Survey (AWCS). The AWCS is a survey fielded to a representative sample of workers in the United States through the American Life Panel (ALP) ran by the RAND Corporation. Two waves of data were collected – one in 2015 and one in 2018 – on the same sample of workers. We construct a balanced panel that consists of all workers participating in both waves. After removing observations with missing values we are left with a sample of 1043 workers. More information on the demographics of our sample can be found in Table 5 in Appendix A.2.

4.1 Description of the Measures

This subsection discusses the measures we use to capture the latent constructs of *work meaning*, *autonomy*, *competence*, *relatedness*, and *beneficence* as defined in section 2. We follow Burbano et al. (2023) in merging our data with variables measuring the pro-social nature of occupations in the O*NET database. An overview of all the measures can be found in Table 1. We re-code these measures such that higher values indicates more work meaning and better scores on the pathway.

Work Meaning. The standard questionnaire to measure work meaning is the Work and Meaning Inventory (WAMI) constructed by Steger et al. (2012). We consider how the measures in the American Working Conditions Survey (AWCS) relate to their conceptualization. The first measure on work meaning asks respondents about whether they have the feeling of doing useful work. This captures the *greater good motivation* and *positive meaning* dimensions discussed in Steger et al. (2012). The same question has been appended to the European Working Conditions Survey (EWCS) and used as a measure of work meaning in both Green and Mostafa (2012) and Nikolova and Cnossen (2020). The other two measures respectively asks respondents whether their work provides them with goals they aspire to, and whether their job provides them with a sense of personal accomplishment. These questions capture the *meaning making through work* dimension, which is important for personal growth.

Autonomy. The measures of autonomy in our data capture control over both the methods of work and over scheduling. To assess control over the work methods, we rely on questions that ask respondents whether (i) they can apply their own ideas in their work, (ii) they are involved in improving the organization and processes of their work, and (iii) they can assess themselves the quality of their work. The final question captures autonomy over the work schedule and asks respondents about their ability to take breaks when they want to. These measures are almost identical to those used in Nikolova and Cnossen (2020) and Burbano et al. (2023).

Competence. We use three questions to measure (feelings of) competence among the respondents in our sample. The first measure is a question that asks respondents whether their job provides them with opportunities to fully use their talents. This captures subjective feelings of competence. The other two measures are again the same as those used in Nikolova and Cnossen (2020). These questions capture a more objective dimension, and ask respondents about whether their job involves complex tasks and if their work confronts them with unforeseen problems which they have to solve on their own. We expect this objective dimension to also be informative about respondents' evaluations regarding their feelings of competence.

Relatedness. The measures of relatedness capture whether respondents feel connected to

Table 1: Measures of Work Meaning and Pathways

Work Meaning (M)	
$Q^{M}_{UsefulWork}$	"You have the feeling of doing useful work". Measured on a five point scale ranging from "Always" to "Never"
$Q^{M}_{GoalsAspirations}$	"Your job provides you with goals to aspire to". Measured on a five point scale ranging from "Always" to "Never".
$Q^{M}_{PersAccomplish}$	"Your job provides you with a sense of personal accomplishment". Measured on a five point scale ranging from "Always" to "Never".
Autonomy (A)	
$Q^{A}_{ApplyOwnIdeas}$	"You are able to apply your own ideas in your work." Measured on a five point scale ranging from "Always" to "Never".
$Q^{A}_{SetSchedule}$	"Can you take breaks when wanted" Measured on a five point scale ranging from "Always" to "Never".
$Q^{A}_{OrgInvolvement}$	"You are involved in improving work organization/processes." Measured on a five point scale ranging from "Always" to "Never".
$Q^{A}_{AssessOwnWork}$	"Does your job involve assessing for yourself the quality of own work?". Measured by a Yes / No indicator.
Competence (C)	
$Q^{C}_{OpportunityTalents}$	"Your job provides you with opportunities to fully use talents"." Measured on a five point scale ranging from "Always" to "Never".
$Q^{C}_{SolveProblems}$	"Generally, does your main paid job involve solving unforeseen problems on your own?" Measured by a Yes / No indicator.
$Q^{C}_{ComplexTasks}$	"Generally, does your main paid job involve complex tasks?" Measured by a Yes / No indicator.
Relatedness (R)	
$Q^{R}_{ManagementAppreciate}$	"Employees are appreciated when they have done a good job". Measured on a five point scale ranging from "Strongly Agree" to "Strongly Disagree".
$Q^{R}_{CooperationColleagues}$	"There is good cooperation between you and your colleagues". Measured on a five point scale ranging from "Strongly Agree" to "Strongly Disagree".
$Q^{R}_{ConflictResolution}$	"Conflicts are resolved fairly" Measured on a five point scale ranging from "Strongly Agree" to "Strongly Disagree".
$Q^{R}_{LikeRespectColleagues}$	"You like and respect your colleagues. Measured on a five point scale ranging from "Strongly Agree" to "Strongly Disagree".
Beneficence (B)	
$Q^{B}_{ImpactSociety}$	"Your work allows you to make a positive impact on society". Measured on a five point scale ranging from "Always" to "Never".
$\mathbf{Q^{B}_{ConcernOthers}}$	"Job requires sensitivity to others' needs and feelings and being understanding and helpful." Measured on a five-point importance scale.
$\mathbf{Q^{B}_{AssistCare}}$	"Providing personal assistance, medical attention, emotional support, or other personal care to others such as coworkers, customers, or patients." Measured on a five-point importance scale."

Notes. Measures of work meaning, autonomy, competence, relatedness, and beneficence in the American Working Conditions Survey (AWCS, waves 2015 and 2018). Combined with occupation-level measures of beneficence from the O*NET database (indicated by bold font) as in Burbano et al. (2023).

their colleagues and to the company. The first set of measures asks respondents whether (i) employees are appreciated when they have done a good job, (ii) there is good cooperation with colleagues, and (iii) they like and respect their colleagues. The final question aims to capture more general relatedness to the work environment, and asks respondents about their beliefs regarding the resolution of conflicts in the workplace. These questions are again comparable to those used in previous work such as Nikolova and Cnossen (2020) and Burbano et al. (2023).

Beneficence. The first measure of benefice asks respondents whether their work allows them to make a positive impact on society. This is a standard question on beneficence that has been used several times (see e.g. Kesternich et al. (2021) or De Schouwer and Kesternich (2023)). The other measures of benefice are introduced in the same way as Burbano et al. (2023), by merging variables from the O*NET database into our data. These variables capture more objective dimensions of beneficence and measure (i) the importance of understanding and helping others, and (ii) providing personal assistance to coworkers, customers, or patients.

Quality of the Measures. We discuss the quality of the measures in more detail in Appendix A.3 but briefly highlight the main results here. The first is that Table 6, which contains information on the distributions of the measures, shows that there is a substantial amount of variation in all measures. We find that even questions that ask respondents about whether they consider their work to be useful contain substantial variation in the answers, and social desirability bias does not seem to be a big problem. We then perform an exploratory factor analysis in Table 7 to further explore the information content of the different measures of the pathways[6] We select four factors based on a scree plot. We find that the exploratory factor model identifies separate factors for relatedness and beneficence, but has trouble distinguishing autonomy from competence. This does not seem to cause problems when we estimate the dedicated measurement system, as seen in Table 8, where we find that all variables are rather informative about the pathway we assigned them to.

[6]We did not include the measures of work meaning in this analysis because all the pathway measures should also measure work meaning to some extent.

5 Results

5.1 The Production Function Estimates

We now discuss the main results obtained from estimating the production function for work meaning. The parameter estimates are presented in Table 2. The first thing we want to note is that accounting for individual-specific (unobserved) heterogeneity is important. In our base specification, the coefficient on autonomy is small and statistically insignificant. This coefficient increases drastically in magnitude and becomes almost significant at the 95% level. An important reason for this is that individuals simply interpret variables such as autonomy very differently, depending on their particular situation. On the other hand, adding occupation fixed effects changes our estimates by only a small margin. This suggests that our model explains the production of meaning in different occupations relatively well. We will thus focus on column (3) of Table 2 from now on, and use these results in subsequent analyses.

The introduction of individual fixed effects raises important concerns related to the use of job changes as the main form of identifying variation – since low levels of work meaning can be an important driver of these changes (see Lavetti (2023) for a related discussion in the context of compensating differentials). This is not a problem in our setting because we find that not a single individual provided the exact same answer in both waves, even among the large sample of respondents that did not change their jobs. This is not surprising, since there is a period of roughly three years between the two survey waves and because the questions on work meaning and the pathways are highly subjective. Respondents' tasks within their current job can change, or even their feelings towards the same set of responsibilities can be different due to other idiosyncratic changes, and this can affect their perceptions of work meaning and the different pathways. This highlights that making work meaningful is something that requires continuous efforts.

The first thing to note is that the direct output elasticities of each pathway – holding the others constant – are all positive and statistically significant, except for autonomy. Autonomy is positive but marginally insignificant at the 5% level. This is in line with the theories proposed in Martela and Riekki (2018) and Cassar and Meier (2018). We find that the direct elasticity of work meaning with respect to competence is the largest at 0.75. This is followed by beneficence at

Table 2: Estimates of the parameters of the production model.

	(1)	**(2)**	**(3)**
Autonomy	-0.041	0.251	0.28
	(-0.098, 0.035)	(-0.029, 0.549)	(-0.032, 0.608)
Beneficence	**0.401**	**0.387**	**0.37**
	(0.291, 0.507)	(0.253, 0.598)	(0.189, 0.598)
Relatedness	**0.126**	0.087	**0.09**
	(0.068, 0.186)	(-0.007, 0.195)	(0.0, 0.199)
Competence	**0.656**	**0.742**	**0.75**
	(0.571, 0.763)	(0.552, 0.877)	(0.552, 0.901)
Autonomy $\times$ Beneficence	-0.094	-0.081	-0.05
	(-0.248, 0.034)	(-0.306, 0.11)	(-0.298, 0.126)
Autonomy $\times$ Relatedness	**0.377**	-0.661	**-0.733**
	(0.142, 0.606)	(-0.989, 0.025)	(-1.045, -0.081)
Autonomy $\times$ Competence	**-0.078**	**0.408**	**0.43**
	(-0.17, -0.004)	(0.123, 0.557)	(0.147, 0.582)
Beneficence $\times$ Relatedness	**0.18**	**-0.178**	-0.158
	(0.072, 0.278)	(-0.336, -0.015)	(-0.348, 0.022)
Beneficence $\times$ Competence	**-0.358**	0.074	0.073
	(-0.48, -0.224)	(-0.112, 0.385)	(-0.117, 0.418)
Relatedness $\times$ Competence	0.055	0.159	0.154
	(-0.032, 0.139)	(-0.032, 0.26)	(-0.046, 0.25)
Number of observations	2086	2086	2086
Number of individuals	1043	1043	1043
Individual FE	No	Yes	Yes
Occupation FE	No	No	Yes

Notes. Point estimates of the production function parameters – the values of γ_P and $\gamma_{PP'}$ from equation (7) – for specifications with and without individual and occupation fixed effects. Below each estimate we present 95% bootstrapped confidence bounds based on 1.000 bootstrapped samples. Bold faced estimates are significant at the 5% level.

0.37 and autonomy at 0.28. The direct output elasticity of relatedness is the smallest at 0.09.

These results are partly in line with previous work such as Martela and Riekki (2018), that also documents relatedness as the pathway with the lowest correlation to work meaning. They do however find only a small correlation for competence, which we document as the main pathway. Other assessments of self-determination theory have nonetheless found a large role for competence (see for example Cerasoli et al. (2016)). These findings are less in line with Nikolova and Cnossen (2020), who finds the largest correlation between work meaning and relatedness. To some extent this likely reflects cultural differences between the United States and Europe, with the former having a work culture more focused on performance. Previous work by Martela and Riekki (2018) also documents significant differences of the effectiveness of different pathways between respondents from India and the United States.[7] We should also note that the direct effect of relatedness decreases by almost 40% upon controlling for individual time-constant heterogeneity in our model, and the role of competence increases. The European Working Conditions Survey (EWCS) and single waves of Amazon Mechanical Turk (Mturk) data from which their estimates are derived do not allow them to control for individual heterogeneity.

We now look at interactions between the different pathways by considering the estimated parameters on the interaction terms. We find that allowing for interactions between the different pathways, and not imposing a prior restriction on their signs, is important. The main result is that we find autonomy and competence to be complementary. This is in line with the long standing theoretical predictions of Ryan and Deci (2000) in social determination theory literature. The interpretation of this interaction straightforward. On the one hand, individuals need sufficient autonomy to be motivated to express their competence, on the other hand, when feelings of competence are low, individuals benefit from more guidance and concrete guidelines. We also document a positive – but smaller and statistically marginally insignificant – complementarity between relatedness and competence. We interpret this interaction in terms of people finding meaning in expressing their competence and skills when working together with people they can relate to.

We also find that some pathways are negatively correlated. The largest negative interaction is between autonomy and relatedness. We believe that the driving force behind this result are

[7] We should note nonetheless that they found the role of competence to be relatively small in the United States.

people on the lower ends of autonomy and relatedness. These individuals have little freedom in deciding whom to work with or take orders from. This can be particularly frustrating when working with individuals that one does not relate to (see Kahane (2017)). On the other hand, individuals with higher levels of autonomy are likely to interact less with their colleagues, and thus would not derive much meaning from more relatedness. The aggregate effect would then balance out to be negative.[8] A negative correlation between autonomy and relatedness is also in line with some influential psychological theories (such as Steinberg and Silverberg (1986)), although not necessarily with the classic self determination theory of Ryan and Deci (2000).

The second relatively large negative interaction is between beneficence and relatedness. To interpret this note that individuals working in jobs with a high pro-social impact usually experience high levels of social interaction – think of nurses, teachers, or therapists.[9] These individuals have a large amount of contact with clients, patients, or students, and for them further connection to their colleagues may not be an important pathway towards meaning. Furthermore these individuals work with other people all day, and too much interpersonal contact can be exhausting. On the other hand, individuals in jobs with a smaller pro-social dimension experience less interpersonal contact while working. Not having colleagues they can relate to can be particularly isolating and harmful for these people. The initial negative sign of this coefficient also hints at a mechanical correlation, which changes upon controlling for time-constant individual heterogeneity. The other interactions between the pathways are small and statistically insignificant.

5.2 Money Metric Benefits

The production function estimates provide us with a useful indication for the direction and significance of the different pathways in generating work meaning. To better understand the magnitude of these results, we translate meaning in monetary terms. We do so by estimating the equilibrium price of work meaning – the *compensating differential* – and then calculate how much dollars changes in different combinations of the pathways generate.

Pricing Work Meaning. To estimate the equilibrium price of work meaning we need to address

[8] Recent work by Zhou (2020) also highlights the dangers associated with higher levels of autonomy, which implies less monitoring, and eventually increases self-interested deviation behavior. We would nonetheless expect this to decrease the level of relatedness too.

[9] These examples draw from Cassar and Meier (2018).

the endogeneity of meaning with respect to wages. The issue is that workers of different levels of productivity divide their total compensation differently between money and meaning, and more productive workers end up with more of both. Because productivity is difficult to control for, naive compensating differentials estimates are biased and typically even 'wrong-signed' (see Hwang et al. (1992) for an overview). To address this problem, we use an estimator recently introduced by Bell (2024).[10] The approach relies on observing an imprecise proxy for ability to shift workers' total compensation.

The Bell (2024) estimator consists of two step. The first stage regresses meaning and monthly wages on an observed ability proxy. In our case the available proxy is years of education. This regression essentially determines the direction in which productivity increases. We then use the predicted values from the first-stage regression as controls in the second stage regression of work meaning on wages. The coefficient on meaning in the second stage regression – ψ_m below – can be interpreted as the compensating differential under the assumption that the proxy variable is (i) informative about total compensation, but (ii) not related to how workers *split* their total compensation into meaning and wages. General measures of productivity that are not specifically manipulated to obtain jobs with different meaning and money combinations – we believe years of education to be general enough[11] – satisfy these assumptions. The regressions that we estimate are:

$$\text{First-Stage:} \quad S_i = \theta_M \ln(M_i) + \theta_W W_i + \xi_i \tag{8}$$

$$\text{Second-Stage:} \quad W_i = \psi_M \ln(M_i) + \psi_S \hat{S}_i + \epsilon_i, \tag{9}$$

where S_i denotes years of schooling, M_i the level of meaning derived from our measurement model, and W_i monthly wages.

The estimated price of meaning can be found in Table 3. We first estimated a simple linear regression of work meaning on wages without the ability proxy as a control (see column (1)). This leads to a small and 'wrong-signed' *positive* compensating differential estimate of 59.94 dollars. This simply tells us that workers with higher levels of productivity obtain higher levels of money

[10] See also Folke and Rickne (2022), Bell et al. (2023), and Burbano et al. (2023).

[11] This proxy is also used inFolke and Rickne (2022), Bell et al. (2023), and Burbano et al. (2023).

and meaning in their jobs. As we can see in column (2), introducing a control for ability does little to address this issue. The final column presents the estimates using the Bell (2024) estimator described above. We find that the equilibrium price of a standard deviation increase in log work meaning is worth 511.98 dollars of monthly wages.

Table 3: The Price of Work Meaning (in dollars)

	Base	Productivity Controls	Bell Proxy
Meaning	59.935 (-59.522 , 179.39)	-27.15 (-137.542, 83.242)	**-511.976** (-807.99 , -228.602)
Partial F			532.626

Notes. Coefficients from regressions of work meaning on monthly wages. The Base specification contains no control variables, the Productivity Controls column introduces years of education as a control for productivity, and the Bell Proxy estimates are obtained as discussed in section 5.2. We report 95% confidence intervals below the estimates, which are derived from T-tests (for the base and productivity specifications) and Anderson-Rubin tests (for the Bell Proxy estimates) as discussed in Andrews et al. (2019) and Bell (2024). Bold faced estimates are significant at the 95% level. The Partial F statistics from the first stage regression is presented in the final row. The First-Stage regression coefficients can be found in Appendix A.5.

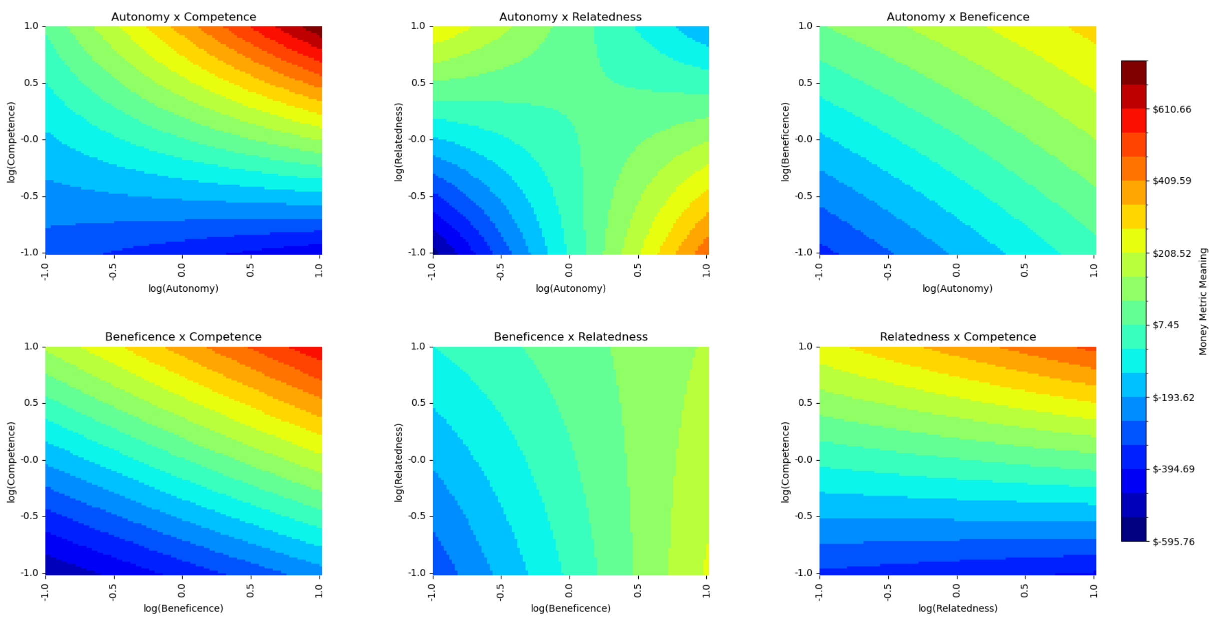
While our compensating differential estimate is hard to compare to previous work, the magnitude is in the range of some *willingness-to-pay* estimates recently reported for beneficence and autonomy. These are different from our estimates because they only reflect worker side preferences and not equilibrium prices, which will differ depending on how these pathways are priced on the demand side of the market. They do nonetheless provide a useful indication about whether our results are in the same the order of magnitude. The most closely related estimates is that of Maestas et al. (2023), who appended a discrete choice experiment to the same American Working Conditions Survey. They find that workers on average value contributing to society at around 181 dollars per month.[12] Similar but slightly higher valuations by workers are reported in Non et al. (2022) and De Schouwer and Kesternich (2024), who estimate values between 150 and 400 euros of monthly wages for increases in beneficence and autonomy in the Netherlands.

Pricing Changes in Pathways. We now compute the value of work meaning – priced using the estimated compensating differentials – for different combinations of the pathways (see Figure 1). The curvature of these level lines informs us about how complementary the two pathways are in generating (monthly dollars worth of) work meaning, with a linear relation indicating the

[12] They estimate a willingness to pay of 3.6% of workers' hourly wage, which we multiply with the average hourly wage and the average number of weekly hours worked in their sample.

absence of any interaction. We want to highlight several results from this figure. First, note that a standard deviation increase in log competence and autonomy would generate more than 600 dollars worth of meaning. This partly due to their immediate positive effects, but these two pathways are also very complementary. The same increase in log beneficence and competence generates a slightly smaller value. The larger direct effect of beneficence on meaning does not make up for the lack of complementarities between these pathways.

Figure 1: Money Metric Meaning for Different Combinations of Pathways



Notes. Monetary values of work meaning for different combinations of the pathways. Derived using the production function estimates for our main specification presented in column (3) of Table 2 and the compensating differential estimates of column (3) of Table 3.

The relation between relatedness and autonomy is more complex because of the presence of a negative complementarity as documented in Table 2. We find that the negative complementarity is particularly relevant at low levels of autonomy and relatedness, as we discussed in the previous section. This could stem from having to work with, or follow order from, individuals that one does not feel related to. The direct positive effect of an increase in either autonomy or competence overshadows the negative complementarity to increase meaning. On the other hand, when levels of relatedness and autonomy are sufficiently high, no additional meaning is generated from a further increasing relatedness. There intuition here is again that individuals with high levels of autonomy are less likely to collaborate intensely with colleagues and thus won't generate a substantial amount of meaning from higher levels of relatedness. The decrease at the upper end is likely just the consequence of our functional form not allowing for more flexible interactions.

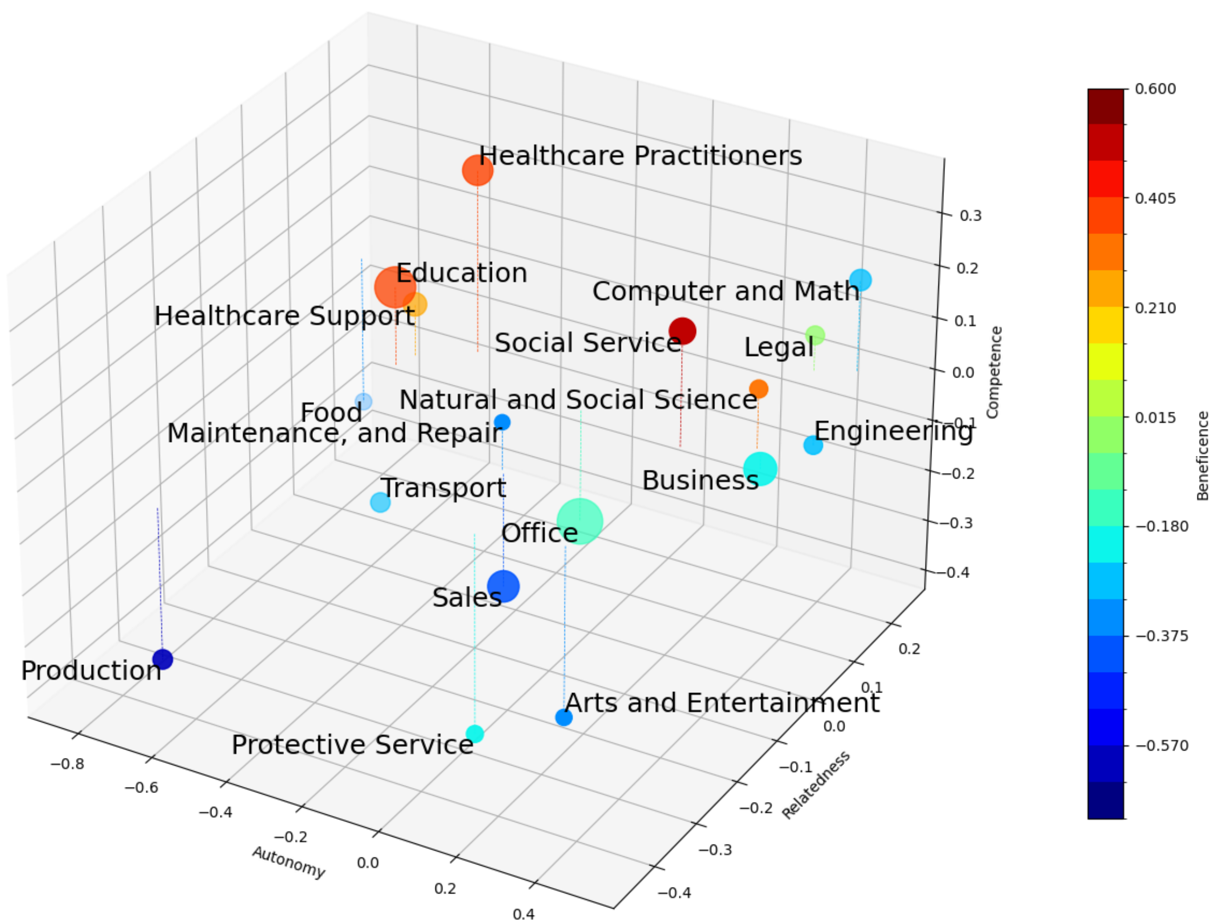
We also find that a standard deviation increasing beneficence results in a roughly 200 dollar gain at any level of relatedness. The negative interaction term here implies that at higher levels of beneficence, an increase in relatedness does not generate any meaning. The intuition here is again that people working jobs with high levels of beneficence often have a large amount of social contact, for example with students or patients, and that these people do derive much meaning from closer connections to or contact with their colleagues. We also find that a joint standard deviation increase in autonomy and beneficence generates almost 400 dollars worth of meaning, with little complementarities, and a joint standard deviation increase in relatedness and competence generates almost 500 dollars.

5.3 Where To Improve?

We now look further into how different occupations score on each pathway, with the aim of identifying where potential improvements can be made. The result from this analysis can be seen in Figure 2. The three dimensional plane in this figure represents occupations' levels of the different dimensions of job design, and the color indicates the level of beneficence. The occupations that are well designed, in terms of providing their workers with feelings of autonomy, relatedness, and competence, are located in the top right corner of the plane. We jointly look at the marginal effects – the elasticities – of work meaning with respect to each pathway in Table 4 to better understand the effectiveness of these potential improvements.

We find that individuals working in technical occupations, such as legal, scientific, and computer and mathematical occupations, and to a lesser extent those in business and engineering, indicate high levels on all dimensions of job design. The main variation is in terms of competence, which individuals in engineering and business indicate to be lower. However, except for scientists, individuals in these jobs indicate average to low levels of beneficence, suggesting that this could be a direction to improve meaning in these jobs. We interpret the efforts that companies have been devoting to corporate social responsibility programs and mission statements as an effort in this direction – see Cassar and Meier (2018) for a broader discussion. When we look at the marginal effects in Table 4 we find that beneficence is also the second most effective pathway towards increasing meaning in these occupations, with elasticities that are around 0.30 on average. The most effective pathway is competence, which is not surprising given our estimates, but this may be harder to improve given that levels of competence are already relatively high in these occupations.

Figure 2: Occupational Scores on Different Pathways



Notes. Scatter plot of average occupational scores on four pathways to work meaning. The size of each marker represents the relative number of individuals working in this occupation in our sample. We omit occupations for which we have less than 30 observations.

The second result we want to highlight from Figure 2 is that individuals in occupations related to healthcare and education report relatively high levels of competence and relatedness, but indicate that their jobs are worse designed in terms of autonomy. These individuals indicate the highest levels of work meaning, except for those in social services. We believe that increasing autonomy could be an important pathway to increase work meaning among these individuals, particularly because of the relatively high levels of competence. We see in Table 4 that autonomy is indeed a relatively effective pathway to meaning with an elasticity that is between 0.35 and 0.40. This is important to highlight, as recent work has documented decreasing levels of autonomy in the healthcare sector related to standardization and routinization of many procedures (see for example Taylor et al. (2002) and Petrakaki and Kornelakis (2016)). We also find that relatedness is a surprisingly effective pathway to increase work meaning in these occupations, with elasticities that similar in magnitude. The main reason for this is the low levels of autonomy, and the negative interaction between autonomy and relatedness. We should note that competence is again the most effective pathway, and that beneficence is also relatively effective, but these may be harder to improve given their high base levels. We also want to point out that these are all large occupations – as indicated by the size of their markers – such that the overall gains in terms of work meaning in the labor force can be substantial.

We finally also want to highlight that there are quite some occupations in which individuals report low levels in all pathways to meaning. The people working in production occupations indicate the lowest values for all pathways, and those in transportation and food-related occupations indicate low values on all aspects except for relatedness, but we found this to be a relatively inefficient pathway to meaning. These results can be related to the findings of Dur and van Lent (2019), who find that a high percentage of individuals working as plant and machine operators, and cooks, waiters or bartenders consider their jobs to be socially useless. These individuals can gain substantial amounts of meaning from an increase in any pathway, as seen in Table 4. We find that relatedness and autonomy are the most effective pathways in production occupations, both with elasticities around 0.50, because of the large cost associated with having both at low levels. Their effect is similar but slightly smaller for individuals working in transport. There are also a couple of occupations bundled in the middle, such as maintenance and repair, office, and sales occupations. An increase in any pathway could also be beneficial

Table 4: Elasticities of Work Meaning – by Occupation

Occupation	$\epsilon_{M,A}$	$\epsilon_{M,B}$	$\epsilon_{M,C}$	$\epsilon_{M,R}$
Protective Service	0.356	0.389	0.702	0.036
Production	0.513	0.450	0.365	0.604
Arts and Entertainment	0.366	0.379	0.784	-0.084
Sales	0.338	0.384	0.680	0.141
Transport	0.234	0.387	0.508	0.457
Engineering	0.237	0.333	0.954	-0.232
Business	0.273	0.346	0.922	-0.200
Food	0.036	0.360	0.391	0.719
Computer and Math	0.193	0.322	0.957	-0.163
Office	0.200	0.356	0.728	0.097
Maintenance, and Repair	0.461	0.406	0.681	0.186
Legal	0.151	0.323	0.936	-0.160
Natural and Social Science	0.302	0.355	0.950	-0.239
Healthcare Support	0.303	0.399	0.561	0.421
Education	0.340	0.409	0.560	0.416
Social Service	0.368	0.377	0.891	-0.140
Healthcare Practitioners	0.370	0.405	0.629	0.351

Notes. This table shows the elasticity of work meaning with respect to each of the different pathways. Computed using the production function estimates from column (3) in Table 2. The occupations are sorted from least to most meaningful, and occupations with less than 30 observations are omitted.

here, but relatedness is less effective because levels of autonomy are not as low.

6 Conclusion

This paper addresses the question of how we can make work more meaningful. We do so by estimating a production function for work meaning, which takes the four pathways identified in theoretical work – autonomy, competence, relatedness, and beneficence – as its inputs (see Martela and Riekki (2018) and Cassar and Meier (2018)). We find that work meaning can be created through all these pathways, but that relatedness may not be the most efficient. We also highlight that there are several important complementarities between the pathways, and find that accounting for individual specific heterogeneity significantly changes the estimates. To better interpret the magnitude of these results, we estimate the compensating differential for work meaning in the labor market, which is priced at more than 500 dollars of monthly wages per standard deviation.

These results highlight both the significance of work meaning, and how effectively it can be created through the different pathways. One interesting direction for future work is to study heterogeneity in the importance of the different pathways in producing work meaning. Since we rely on fixed effects estimation, we could not study differences in the effectiveness of the pathways by variables such as gender or education. Several recent papers have highlighted that women sort into jobs with higher levels of beneficence (see Burbano et al. (2023) or De Schouwer and Kesternich (2023)), which may be explained by beneficence being a more efficient pathway to meaning for women than it is for men. Another direction is to extend the production model, for example by modeling effort as an additional input (as suggested in Cassar and Meier (2018)), and incorporating it further into models of labor supply. Given the magnitude of our compensating differential estimates, this could have substantial implications.

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A Appendix

A.1 Model Equations

We present the omitted equations from the main text below. Note that substituting the residual measures defined by equation (6) into our production function defined by equation (2) yields:

$$\hat{M}_{ij} - \tilde{\psi}_{ij}^{M} = \Theta_i + \sum_{P \in \mathcal{P}} \gamma_P (\hat{P}_{ij} - \tilde{\psi}_{ij}^{P}) + \sum_{P \in \mathcal{P}} \sum_{\substack{P' \in \mathcal{P} \\ P' \neq P}} \gamma_{PP'} (\hat{P}_{ij} - \tilde{\psi}_{ij}^{P})(\hat{P'}_{ij} - \tilde{\psi}_{ij}^{P'}) + \eta_{ij},$$

where $\tilde{\psi}_{ij}^{F} = \psi_{ij}^{F} / \lambda_j^{F}$, for $F \in \{\mathcal{P}, M\}$. We re-arrange this into equation (7) of the main text, with the equation for the error term (ξ_{ij}) then being:

$$\xi_{ij} = \eta_{ij} + \tilde{\psi}_{ij}^{M} - \Big[\sum_{P \in \mathcal{P}} \gamma_P \tilde{\psi}_{ij}^{P} \Big] + \sum_{P \in \mathcal{P}} \sum_{\substack{P' \in \mathcal{P} \\ P' \neq P}} \gamma_{PP'} \Big[\tilde{\psi}_{ij}^{P} \cdot \tilde{\psi}_{ij}^{P'} - \hat{P}_i \cdot \tilde{\psi}_{ij}^{P} - \hat{P}'_i \cdot \tilde{\psi}_{ij}^{P} \Big]. \quad (10)$$

A.2 Sample Summary Statistics

This Appendix discusses the main demographics for our sample. These are the demographics during the first wave (2015). We find that the average age in our sample is relatively high at 46.96 years old. People are relatively highly educated, with on average just about a bachelor's degree. We also have slightly more women than men in our sample. On average, people work a full-time position with 40 hours per week, but a substantial fraction of the workers in our sample work relatively long hours (more than 45 per week). The mean gross monthly income is roughly 5900 dollars. Note that when estimating the compensating differentials, we remove outliers in wages. To make the sample representative for the United States population, a set of sample weights is provided in Maestas et al. (2017). These weights are not yet used in our analysis.

Table 5: Sample Summary Statistics.

Variable	mean	std	min	25%	50%	75%	max
Age	46.96	11.53	18	38	49	56	70
Years of Education	12.00	2.03	4	10	13	13	16
Male	0.41	0.49	0	0	0	1	1
Monthly Income	5877.85	29330.28	0	2250	3833.33	6000	916666.67
Weekly Hours Worked	40.04	10.62	1	38	40	45	112
Observations	1,043						

Notes. Summary statistics – main sample.

A.3 The Measures

A.3.1 Descriptive Statistics

We briefly discusses the distributions of our measures. The main takeaway that can be made from Table 6 is that we generally have a substantial amount of variation in all our measures. This is true even for measures regarding the usefulness of individuals' jobs, where the standard deviation is about one and a substantial fraction of more than 25% of respondents indicates rather low levels of usefulness. The same is true for the measures of impact on society. We find that the fraction of individuals reporting only a minimal positive impact on society is more than 10%. This variation is important, and addresses the concern that everyone may find their own work to be extremely useful or beneficial. We should also note that even our binary measures, such as whether an individual can assess the quality of their own work or whether their job features solving complex tasks, contain a reasonable amount of variation, as only 83% (and 73%) of individuals respond positively.

A.3.2 Exploratory Factor Analysis

We now estimate an Exploratory Factor Analysis (EFA) to explore how the different measures are interrelated. The results from this analysis can be seen in Table 7. We select four factors based on a scree plot and our theoretical predictions. We find that the first factor captures relatedness, with all measures that we labeled as capturing relatedness having loadings between 0.6 and 0.8. The second factor captures beneficence, with two measures having loadings over 0.9 and the third measure having the third highest loading at 0.17. The third factor captures both autonomy and competence, and the fourth capturing competence and beneficence. This is important to

Table 6: Distributions of the Measures.

	Mean	Std. Dev.	10%	25 %	50 %	75 %	90 %
Work Meaning (M)							
$Q^M_{UsefulWork}$	2.81	0.99	2.00	2.00	3.00	4.00	4.00
$Q^M_{GoalsAspirations}$	2.41	1.13	1.00	2.00	3.00	3.00	4.00
$Q^M_{PersAccomplish}$	2.71	1.00	1.00	2.00	3.00	3.00	4.00
Autonomy (A)							
$Q^A_{ApplyOwnIdeas}$	2.62	1.04	1.00	2.00	3.00	3.00	4.00
$Q^A_{SetSchedule}$	2.60	1.24	1.00	2.00	3.00	4.00	4.00
$Q^A_{OrgInvolvement}$	2.34	1.19	1.00	2.00	2.00	3.00	4.00
$Q^A_{AssessOwnWork}$	0.83	0.38	0.00	1.00	1.00	1.00	1.00
Competence (C)							
$Q^C_{OpportunityTalents}$	2.50	1.07	1.00	2.00	3.00	3.00	4.00
$Q^C_{SolveProblems}$	0.85	0.36	0.00	1.00	1.00	1.00	1.00
$Q^C_{ComplexTasks}$	0.73	0.44	0.00	0.00	1.00	1.00	1.00
Relatedness (R)							
$Q^R_{ManagementAppreciates}$	2.65	1.10	1.00	2.00	3.00	3.00	4.00
$Q^R_{CooperationColleagues}$	3.04	0.86	2.00	3.00	3.00	4.00	4.00
$Q^R_{ConflictResolution}$	2.60	1.02	1.00	2.00	3.00	3.00	4.00
$Q^R_{LikeRespectColleagues}$	3.09	0.80	2.00	3.00	3.00	4.00	4.00
Beneficence (C)							
$Q^B_{ImpactSociety}$	2.49	1.19	1.00	2.00	3.00	3.00	4.00
$Q^B_{ConcernOthers}$	3.93	0.35	3.50	3.77	3.86	4.14	4.51
$Q^B_{AssistCare}$	2.93	0.65	2.33	2.59	2.76	2.99	4.31

Notes. Distribution of the different measures of work meaning, autonomy, competence, and beneficence used in the main analysis.

take into account when we looking at our dedicated measurement system.

Table 7: Exploratory Factor Analysis.

Measure	Factor 1	Factor 2	Factor 3	Factor 4
Autonomy (A)				
$Q^{A}_{ApplyOwnIdeas}$	0.159	-0.076	**0.627**	0.256
$Q^{A}_{SetSchedule}$	0.104	-0.165	**0.41**	0.068
$Q^{A}_{OrgInvolvement}$	0.192	-0.035	**0.601**	0.247
$Q^{A}_{AssessOwnWork}$	0.059	0.046	**0.392**	0.031
Competence (C)				
$Q^{C}_{OpportunityTalents}$	0.283	0.042	0.209	**0.768**
$Q^{C}_{SolveProblems}$	-0.012	0.052	**0.492**	-0.032
$Q^{C}_{ComplexTasks}$	-0.004	0.021	**0.461**	0.046
Relatedness (R)				
$Q^{R}_{ManagementAppreciate}$	**0.634**	-0.038	0.113	0.341
$Q^{R}_{CooperationColleagues}$	**0.817**	0.026	0.113	0.07
$Q^{R}_{ConflictResolution}$	**0.697**	-0.049	0.096	0.26
$Q^{R}_{LikeRespectColleagues}$	**0.77**	0.022	0.078	0.095
Benefice (C)				
$Q^{B}_{ImpactSociety}$	0.264	0.177	0.138	**0.642**
$Q^{B}_{ConcernOthers}$	-0.003	**0.963**	-0.008	0.086
$Q^{B}_{AssistCare}$	-0.005	**0.917**	-0.023	0.086

Notes. Exploratory factor analysis on the different pathways to work meaning. The number of factors is imposed based on theoretical predictions and a scree plot.

A.4 Measurement System Estimates

We now look at the results of the dedicated measurement system in Table 8. This shows that all measures load in the expected direction, and that the weights for the different measures are relatively comparable. This means that the information content about the latent pathways is similar for all measures. We find that the least informative measure for its pathway is the autonomy variable that captures whether an individual can assess the quality of their own work. But this variable is still relatively informative and is binary, so we would expect this to contain somewhat less information.

Table 8: Measurement System Estimates.

Measure	weights	intercepts
Work Meaning (M)		
$Q^{M}_{UsefulWork}$	1.000	2.810
$Q^{M}_{GoalsAspirations}$	1.034	2.411
$Q^{M}_{PersAccomplish}$	1.041	2.709
Autonomy (A)		
$Q^{A}_{ApplyOwnIdeas}$	1.000	2.618
$Q^{A}_{SetSchedule}$	0.802	2.599
$Q^{A}_{OrgInvolvement}$	0.996	2.338
$Q^{A}_{AssessOwnWork}$	0.156	0.830
Competence (C)		
$Q^{C}_{OpportunityTalents}$	1.000	2.504
$Q^{C}_{SolveProblems}$	0.772	0.852
$Q^{C}_{ComplexTasks}$	1.135	0.734
Relatedness (R)		
$Q^{R}_{ManagementAppreciate}$	1.000	2.647
$Q^{R}_{CooperationColleagues}$	0.892	3.043
$Q^{R}_{ConflictResolution}$	1.011	2.596
$Q^{R}_{LikeRespectColleagues}$	0.820	3.090
Benefice (C)		
$Q^{B}_{ImpactSociety}$	1.000	2.486
$Q^{B}_{ConcernOthers}$	1.208	3.934
$Q^{B}_{AssistCare}$	1.460	2.925

Notes. Results from estimating the measurement system in equations 4 and 5.

A.5 The Bell Estimator

This appendix describes the first-stage results from the estimator by Bell (2024) discussed in section 5.2. We use years of education as the productivity proxy as mentioned in the main text. To price the amenities we removed 1% outliers in wages at the bottom and 5% at the top. We ran the regression on both waves combined but found similar results when running this on either wave separately. While this regression has no structural interpretation, positive signs on the coefficient reflect that these are forms of compensation that workers enjoy – as discussed in Bell (2024). We find this to be the case for both work meaning and wages, as seen in Table 9.

Table 9: Distributions of the Measures.

	Years of Education
Cons.	**10.64**
	(0.08)
Work Meaning	**0.15**
	(0.04)
Monthly Wages (in 1.000$)	**0.30**
	(0.02)
R^2	0.16
Adj. R^2	0.15
Num. obs.	1944

Notes. First stage regression of Bell (2024) estimator. Standard errors in parentheses. Bold-faced estimates are significant at the 5% level. Wages are expressed in thousand dollar per month.

A.6 The Estimation Procedure

We first define $\mathcal{S}$ as the union of the set of all pathways and the set of all pairwise combinations of pathways:

$$\mathcal{S} = \mathcal{P} \cup \{xy \mid x, y \in \mathcal{P}, x \neq y\} \tag{11}$$

The first-stage regression we run are of the form:

$$\hat{P}_j^{FS} = \beta_P \hat{P}_k + \sum_{\substack{P' \in \mathcal{S} \\ P' \neq P}} \beta_P \hat{P}_j' + \nu_j \text{ for all } P \in \mathcal{S} \text{ and with } j \neq k, \tag{12}$$

The second stage regressions are then:

$$\hat{M}_j = \Theta + \sum_{P \in \mathcal{P}} \gamma_P \hat{P}_j^{FS'} + \sum_{P \in \mathcal{P}} \sum_{\substack{P' \in \mathcal{P} \\ P' \neq P}} \gamma_{PP'} \hat{P_j P_j'}^{FS} + \epsilon_j. \tag{13}$$